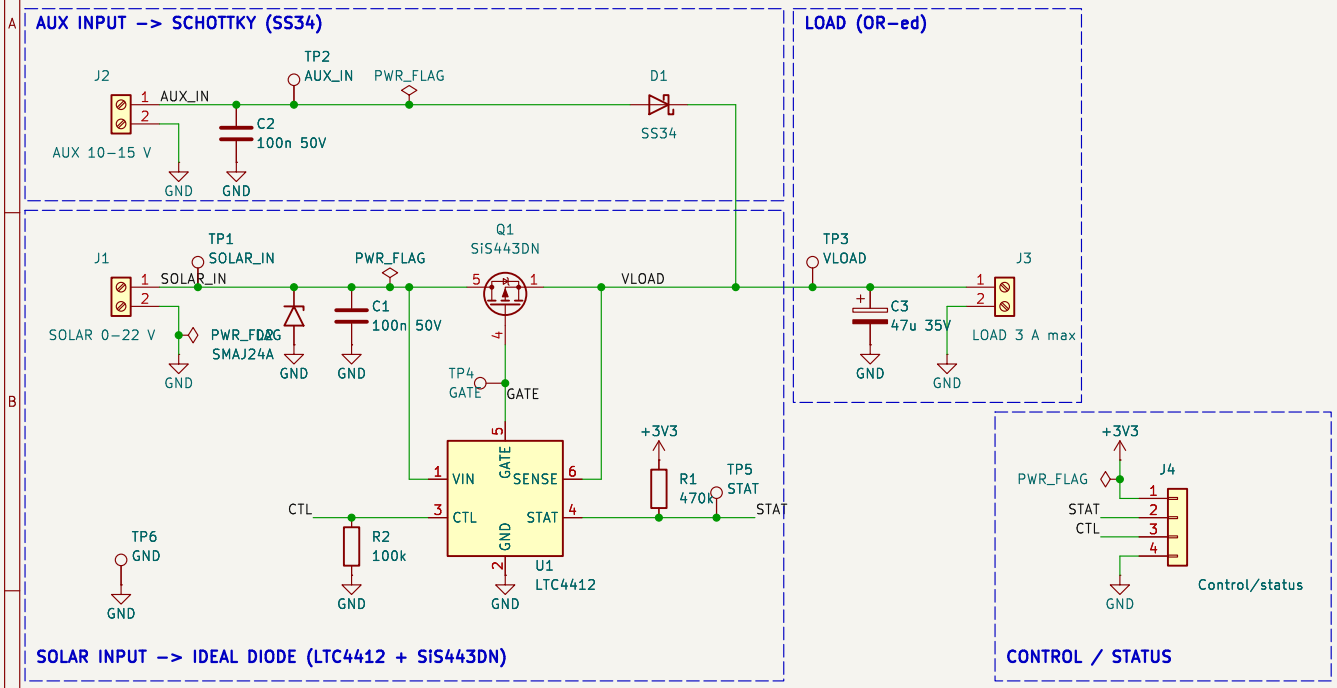
WARNING: Reference design. Test first with a current-limited supply and protection. Not for unattended use.

Design only, not fabricated, not bench-tested.



KEY VALUES (see README for derivations and sources)

Ideal diode: LTC4412 holds VIN - VSENSE at VFR = 10-32 mV until Q1 is fully on; then drop = I x RDS(on).

Q1 SiS443DN: RDS(on) <= 16 mOhm at VGS = -4.5 V. LTC4412 gate clamp 6.3-7.7 V (VIN >= 6.3 V). 3 A -> <= 48 mV, <= 0.14 W.

D1 SS34: VF <= 0.5 V at 3 A, 25 C -> <= 1.5 W at 3 A, about 82 K rise (55 C/W). IR <= 0.5 mA (25 C), 20 mA (100 C).

Solar input: LTC4412 VIN operating 2.5-28 V, abs max 36 V. Panel Voc must stay <= 22 V. D2 SMAJ24A: VR 24 V, VC 38.9 V @ 10.3 A (> 36 V!).

STAT sinks 6-17 uA when the solar path is off (VLOAD > SOLAR_IN + 10..32 mV). R1 470k -> high >= 2.8 V, low <= 0.5 V at 3.3 V.

CTL high (>= 0.9 V) forces Q1 off (e.g. to read panel Voc). R2 keeps it low if the header is open.

C3: LTC4412 suggests 1-47 uF on the load; electrolytic ESR damps the MLCC hot-plug ringing the datasheet warns about.

No load fuse and no panel reverse-polarity protection: use a keyed panel connector (see README Known limitations).

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Abdulazeez Bello

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Title: Solar OR-ing: LTC4412 ideal diode vs Schottky

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